

SICP Setup: MIT Scheme + VS Code

1. Install MIT Scheme

macOS

```
bash  
  
brew install mit-scheme
```

Ubuntu / Debian

```
bash  
  
sudo apt update  
sudo apt install mit-scheme
```

Red Hat / Fedora / CentOS

```
bash  
  
sudo dnf install mit-scheme
```

openSUSE

```
bash  
  
sudo zypper install mit-scheme
```

Verify:

```
bash  
  
mit-scheme --version
```

2. Set Up VS Code

Install the extension

Install **vscode-scheme** for syntax highlighting:

- Open VS Code
- `Cmd/Ctrl+Shift+X` → search `vscode-scheme`
- Install `sjhuangx.vscode-scheme`

Add the keybinding

`Cmd/Ctrl+Shift+P` → **Open Keyboard Shortcuts (JSON)** — add:

```
json
[
  {
    "key": "ctrl+enter",
    "command": "workbench.action.terminal.runSelectedText"
  }
]
```

`Ctrl+Enter` works on both Mac and Linux.

3. Running Code

Interactive REPL — send selected expression

1. Open the VS Code terminal: `Ctrl+``
2. Start the REPL:

```
bash
mit-scheme
```

3. Wait for the `[ ]=>` prompt
4. In your `.scm` file, select any expression
5. Hit `Ctrl+Enter` — it evaluates in the running REPL

Example — select and send:

```
scheme
(define (square x) (* x x))
(square 5)           ; => 25
```

Load a whole file into the REPL

With `mit-scheme` running in the terminal:

```
scheme
(load "scratch.scm")
```

This loads and evaluates the entire file, leaving you in the REPL to interact with everything defined in it.

Run/interpret an entire file non-interactively

```
bash
```

```
mit-scheme --quiet < scratch.scm
```

Interprets the file top to bottom and exits. Use this to check output without an interactive session.